

Question				Marking details			Marks Available					
							AO1	AO2	AO3	Total	Maths	Prac
4 FT	(a)	(i)		$^{141}_{56}\text{Ba} (2) + ^{92}_{36}\text{Kr} + 3\ ^1_0\text{n} (1)$				3		3	3	
		(ii)		Fission , a moderator , neutrons , control rods (1) × 4			4			4		
		(iii)		Uranium isotope	Number of protons	Number of neutrons		2		2	2	
			uranium-235	92	143							
			uranium-236	92	144 (1)							
	(b)	(i)		3 rd and 4 th statements correct: The time taken for the number of undecayed nuclei to halve ✓(1) The time taken for the activity to halve ✓(1)			2			2		
		(ii)		One half life (1) So ½ (1) ½ alone as answer (2)			1	1		2		
		(iii)		U-235 will have gone through more than 6 half lives / U-235 decays quicker (1) Therefore {fraction/amount} (of U-235) must be smaller so don't agree (1) OR U-238 has only gone through 1 half life /U-238 decays slower (1) Therefore {fraction/amount} of (of U-238) must be larger so don't agree (1)					2	2		
				Question 4 total 15			7	6	2	15	5	